PERCENTAGE YIELD

1 Iron is made by reduction of iron oxide with carbon monoxide. $Fe_2O_3 + 3\ CO \rightarrow 2\ Fe + 3\ CO_2$

a) Calculate the mass of iron that can be formed from 126 g of iron oxide.

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b) 78.5 g of iron was formed in this reaction. Calculate the percentage yield.

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2 Chlorine can be made by the electrolysis of sodium chloride solution. $2\ NaCl + 2\ H_2O \rightarrow 2\ NaOH + Cl_2 + H_2$

a) Calculate the mass of chlorine that can be formed from 50.0 g of sodium chloride.

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b) 25.0 g of chlorine was formed in this reaction. Calculate the percentage yield.

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3 Chromium is a useful metal. It is extracted from chromium oxide by reaction with aluminium. $Cr_2O_3 + 2\ Al \rightarrow 2\ Cr + Al_2O_3$

a) Calculate the mass of chromium that can be formed from 1.25 kg of chromium oxide.

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b) 756 g of chromium was formed in this reaction. Calculate the percentage yield.

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4 Ammonia is made by reacting hydrogen with nitrogen. $3\ H_2 + N_2 \rightarrow 2\ NH_3$

a) Calculate the mass of ammonia that can be formed from 12.0 g of hydrogen.

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b) 20.3 g of ammonia was formed in this reaction. Calculate the percentage yield.

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c) Give three potential reasons why the percentage yield was less than 100%.

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2 - ……

3 - ……

5 Titanium is made by the reaction of titanium chloride with sodium. $TiCl_4 + 4\ Na \rightarrow Ti + 4\ NaCl$

a) Calculate the mass of titanium that can be formed from 10.0 kg of titanium chloride.

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c) 1950 g of titanium was formed in this reaction. Calculate the percentage yield.

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Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Can find moles from mass			Gives units when appropriate		
Shows suitable working			Can use reacting ratios in equations			Suitable rounding		
Can work out M_r			Can find mass from moles			Can convert units		
Which numbers are in formula			Can work out % yield			Understands why yield is < 100%		